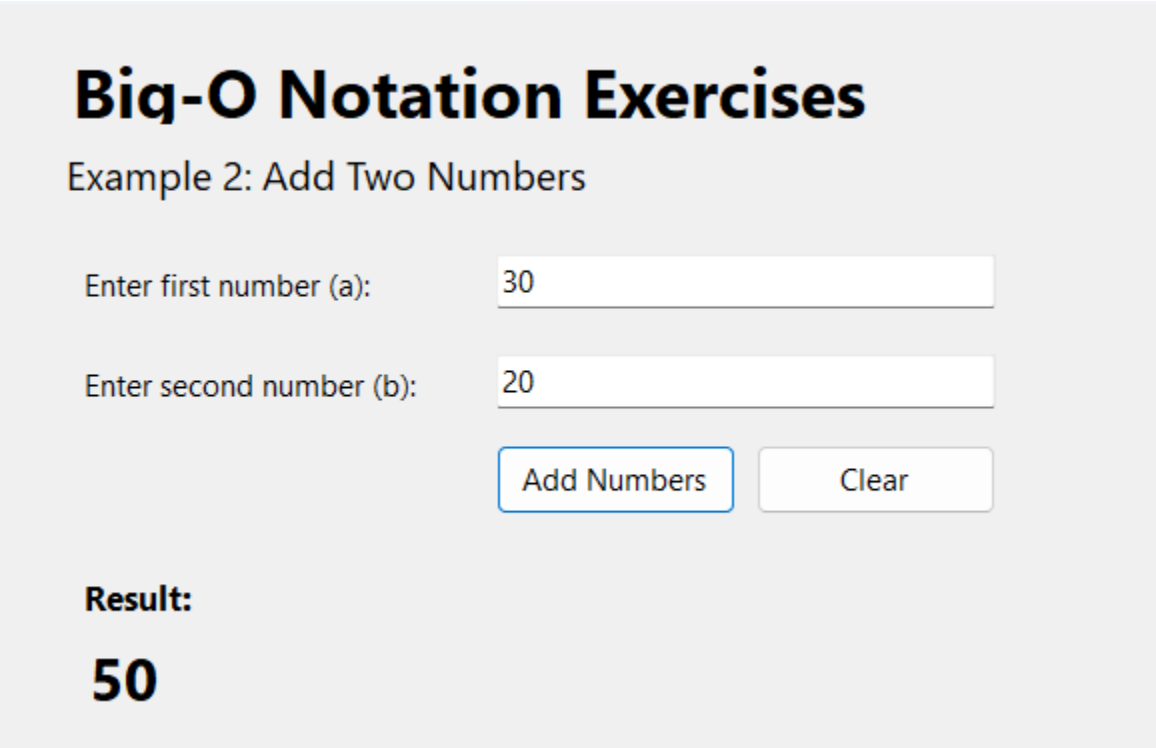
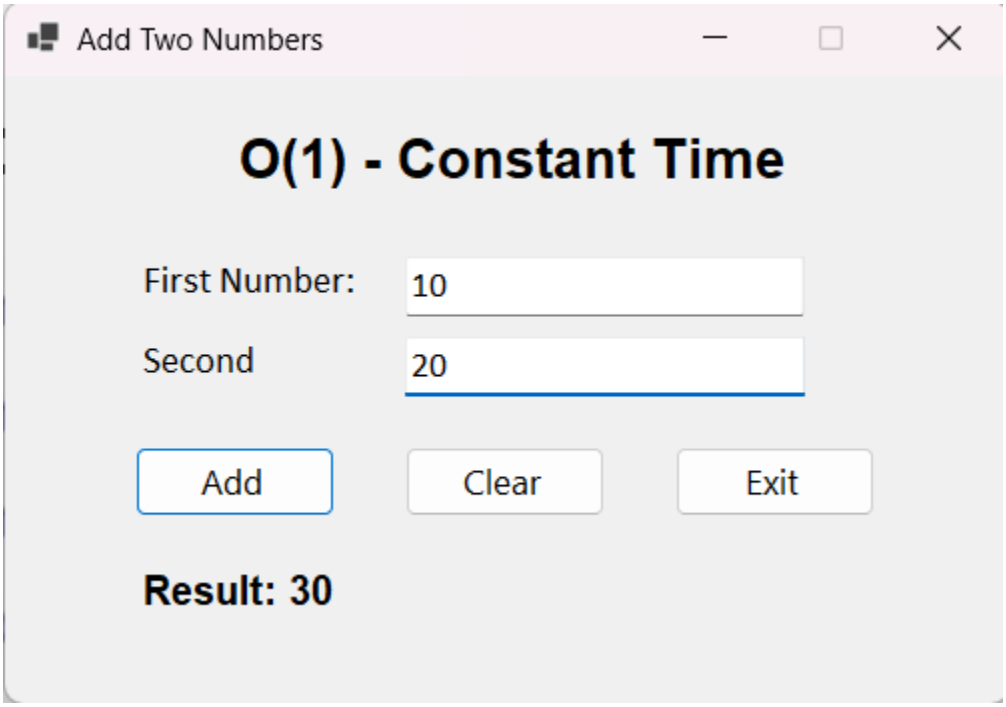
Constant time

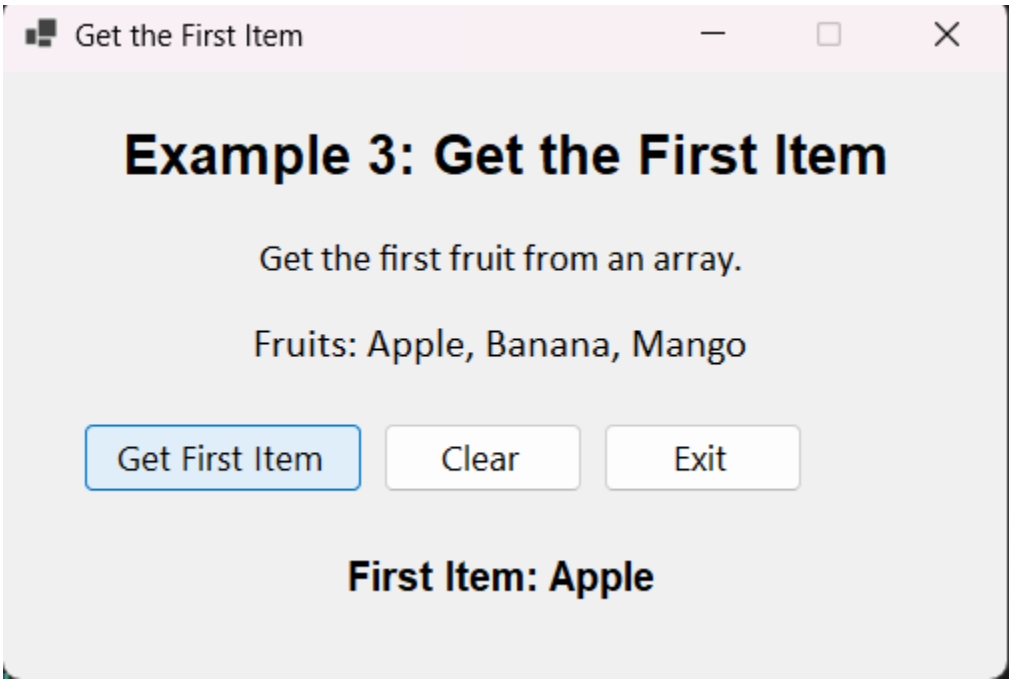
Example1



Example2

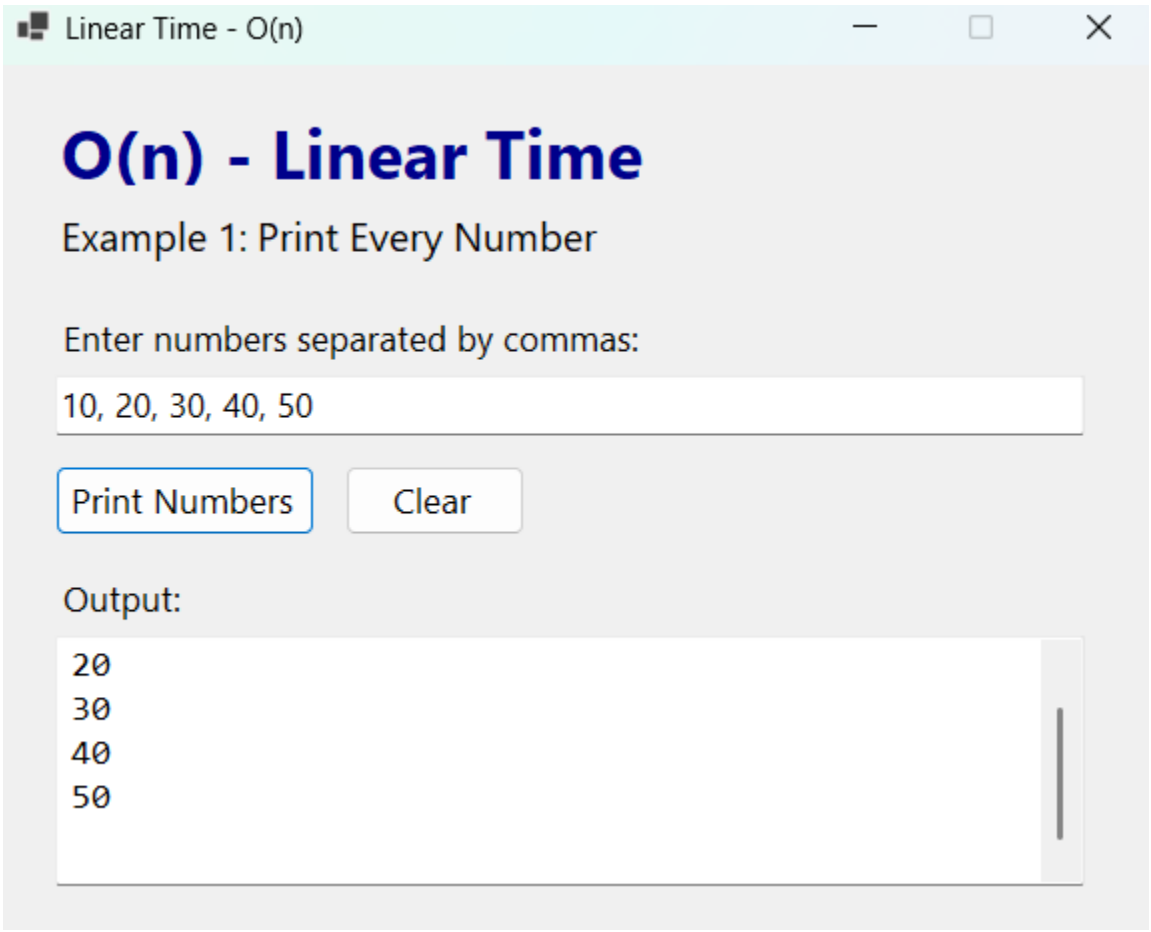


Example3

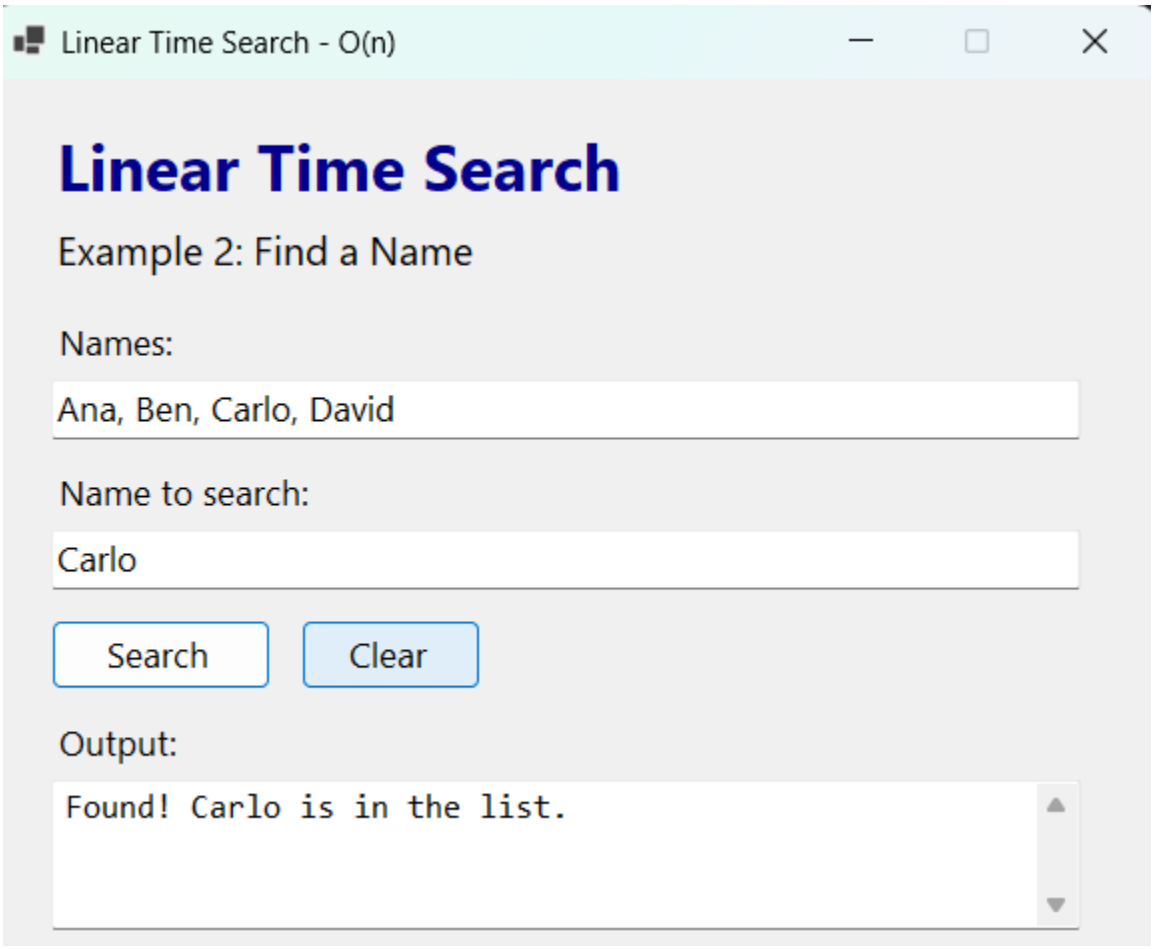


Linear time

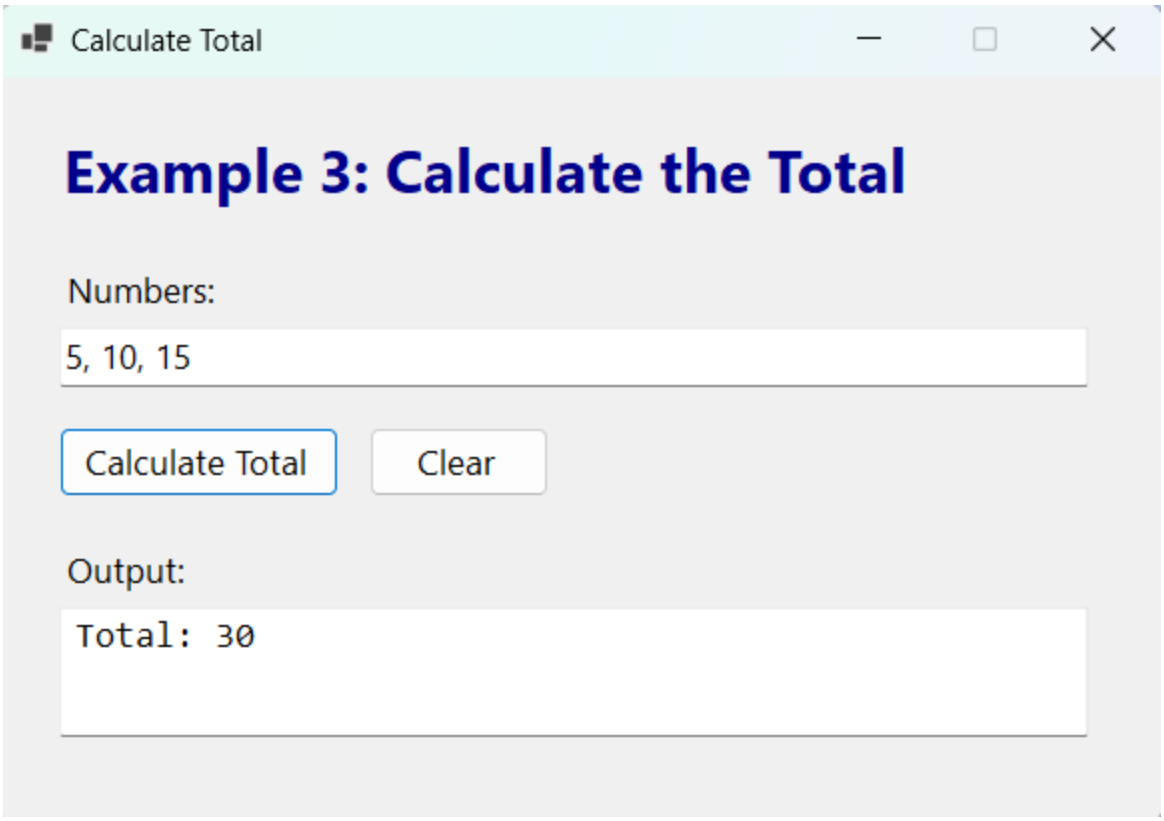
Example1


Linear Time - O(n)
O(n) - Linear Time
Example 1: Print Every Number
Enter numbers separated by commas:
10, 20, 30, 40, 50
Print Numbers
Clear
Output:
20
30
40
50

Example2

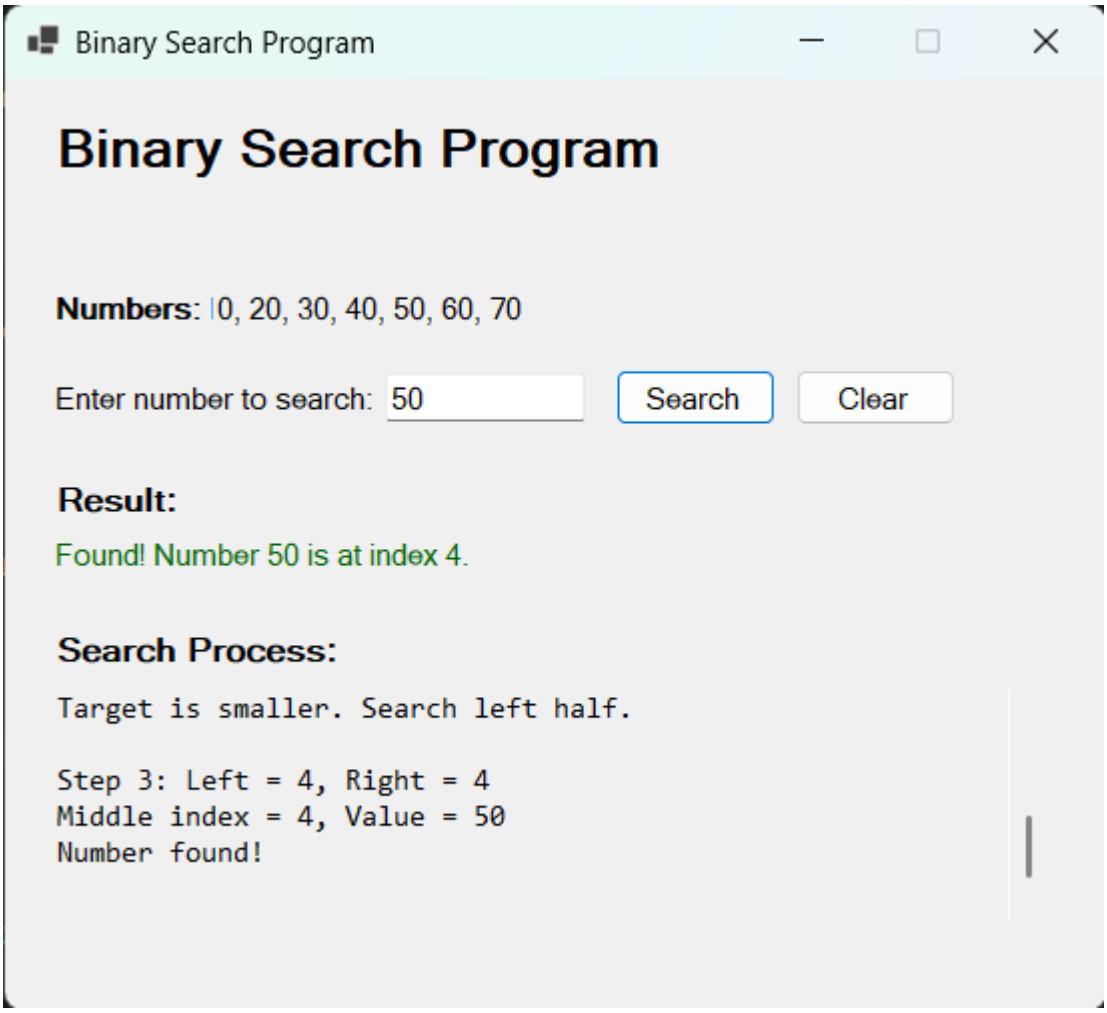


Example3



Logarithmic time

Example1


Binary Search Program
Binary Search Program
Numbers: 0, 20, 30, 40, 50, 60, 70
Enter number to search: 50
Search
Clear
Result:
Found! Number 50 is at index 4.
Search Process:
Target is smaller. Search left half.
Step 3: Left = 4, Right = 4
Middle index = 4, Value = 50
Number found!

Exampl2

Example 2: Keep Dividing by 2

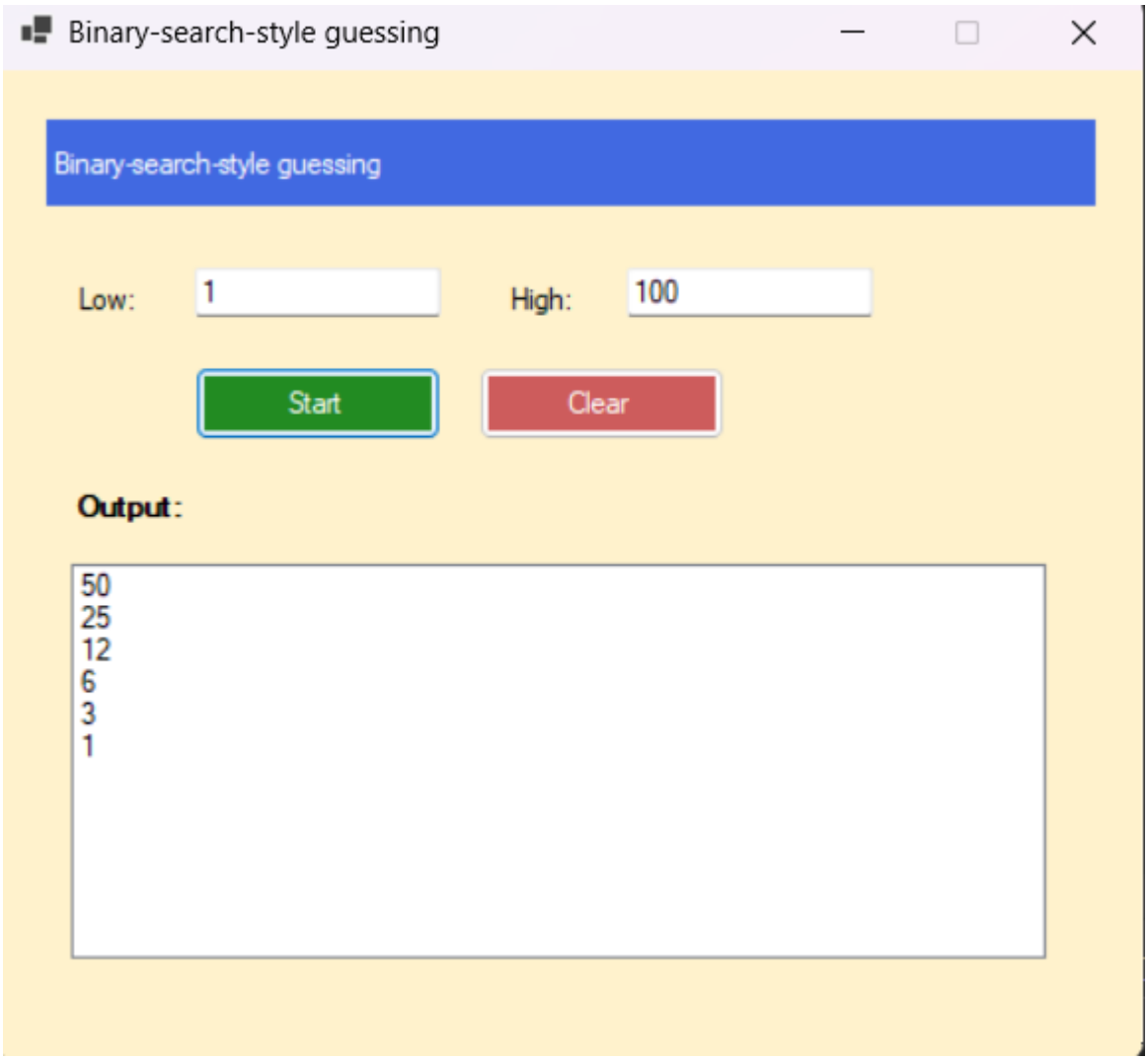
Enter a number: 16

Start Clear

Output:

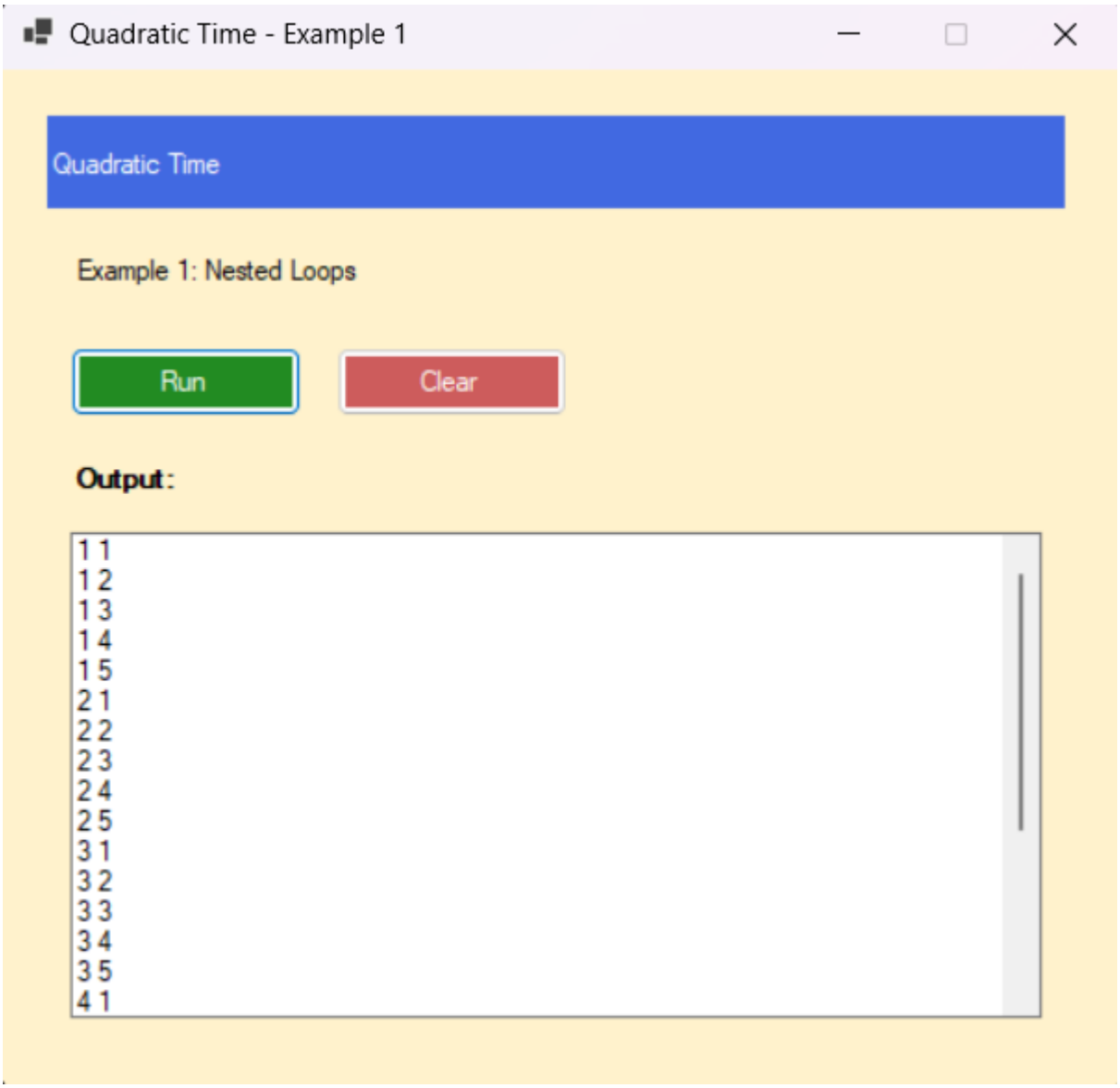
8
4
2
1

Example3

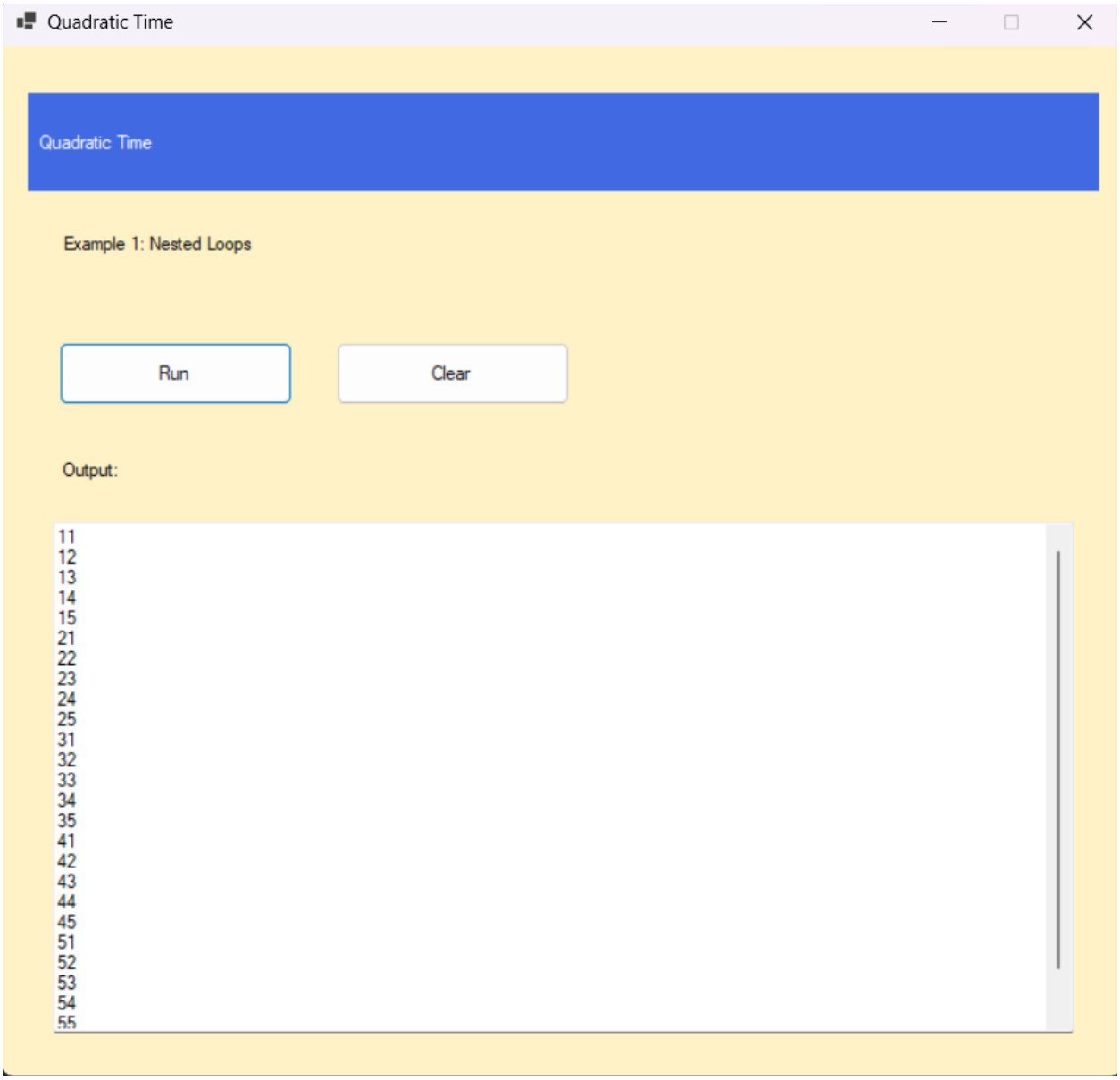


Quadratic time

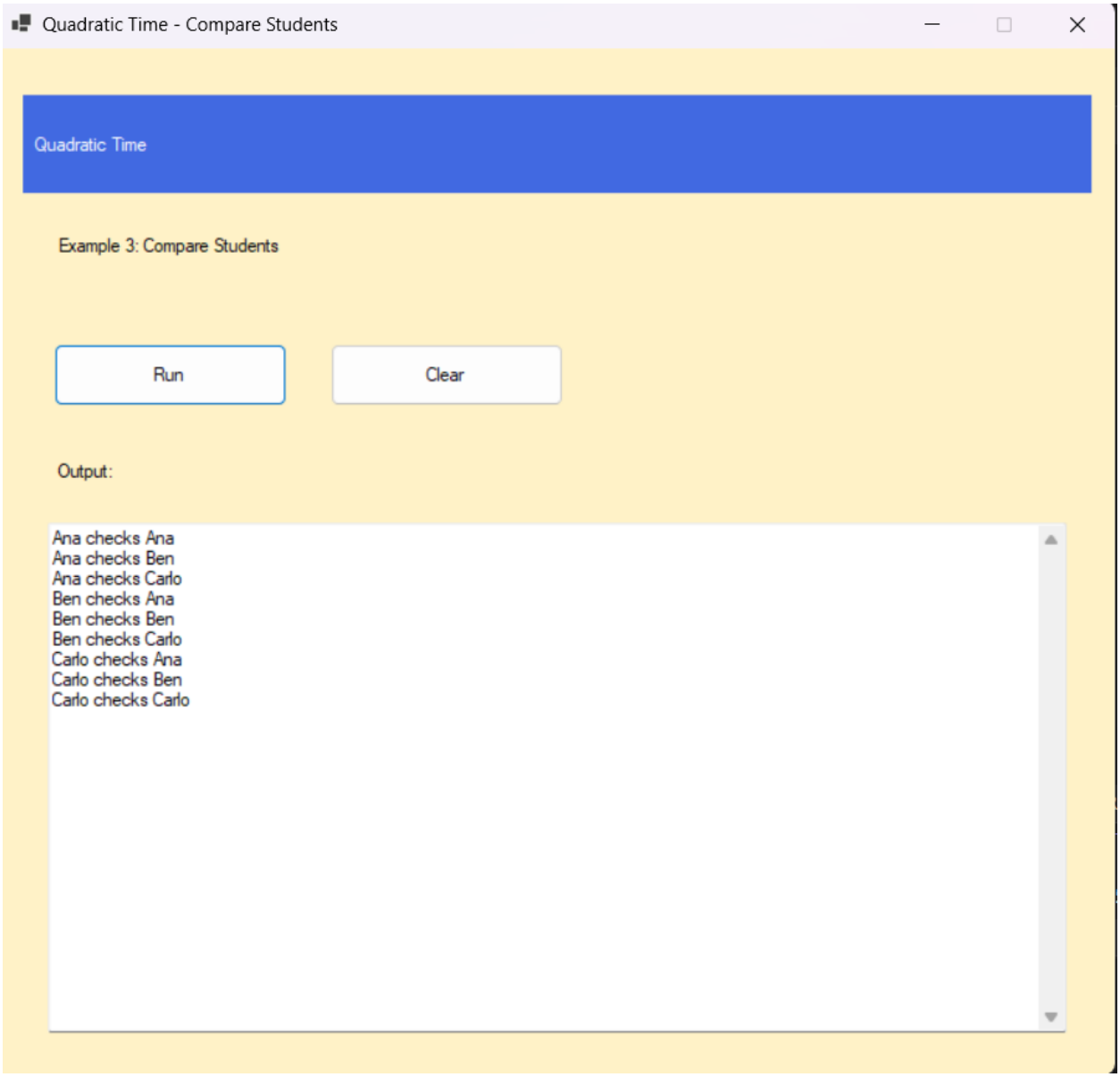
Example1


Quadratic Time - Example 1
Quadratic Time
Example 1: Nested Loops
Run
Clear
Output:
1 1
1 2
1 3
1 4
1 5
2 1
2 2
2 3
2 4
2 5
3 1
3 2
3 3
3 4
3 5
4 1

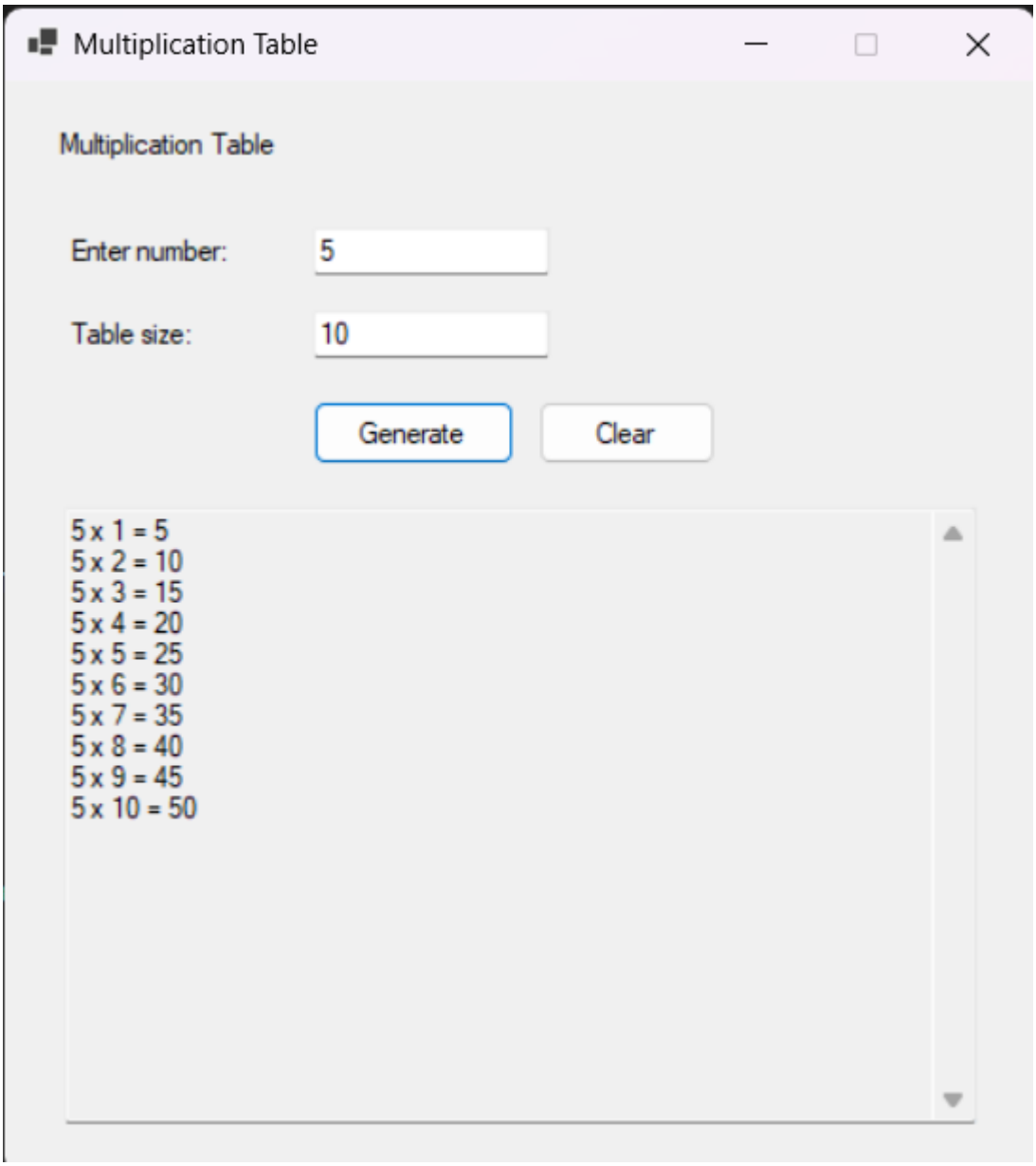
Example2


Quadratic Time
Quadratic Time
Example 1: Nested Loops
Run
Clear
Output:
11
12
13
14
15
21
22
23
24
25
31
32
33
34
35
41
42
43
44
45
51
52
53
54
55

Example3

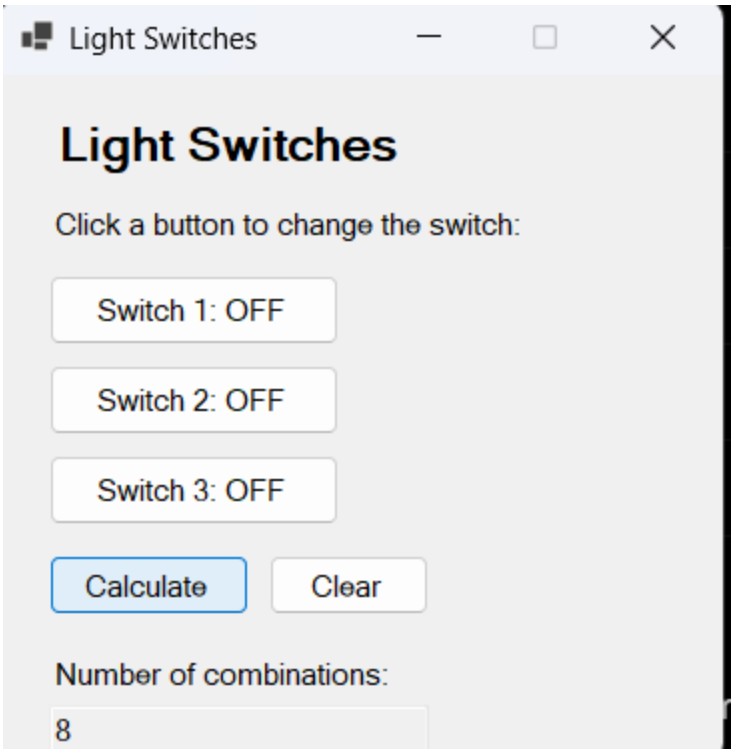

Quadratic Time - Compare Students
Quadratic Time
Example 3: Compare Students
Run
Clear
Output:
Ana checks Ana
Ana checks Ben
Ana checks Carlo
Ben checks Ana
Ben checks Ben
Ben checks Carlo
Carlo checks Ana
Carlo checks Ben
Carlo checks Carlo

Example4

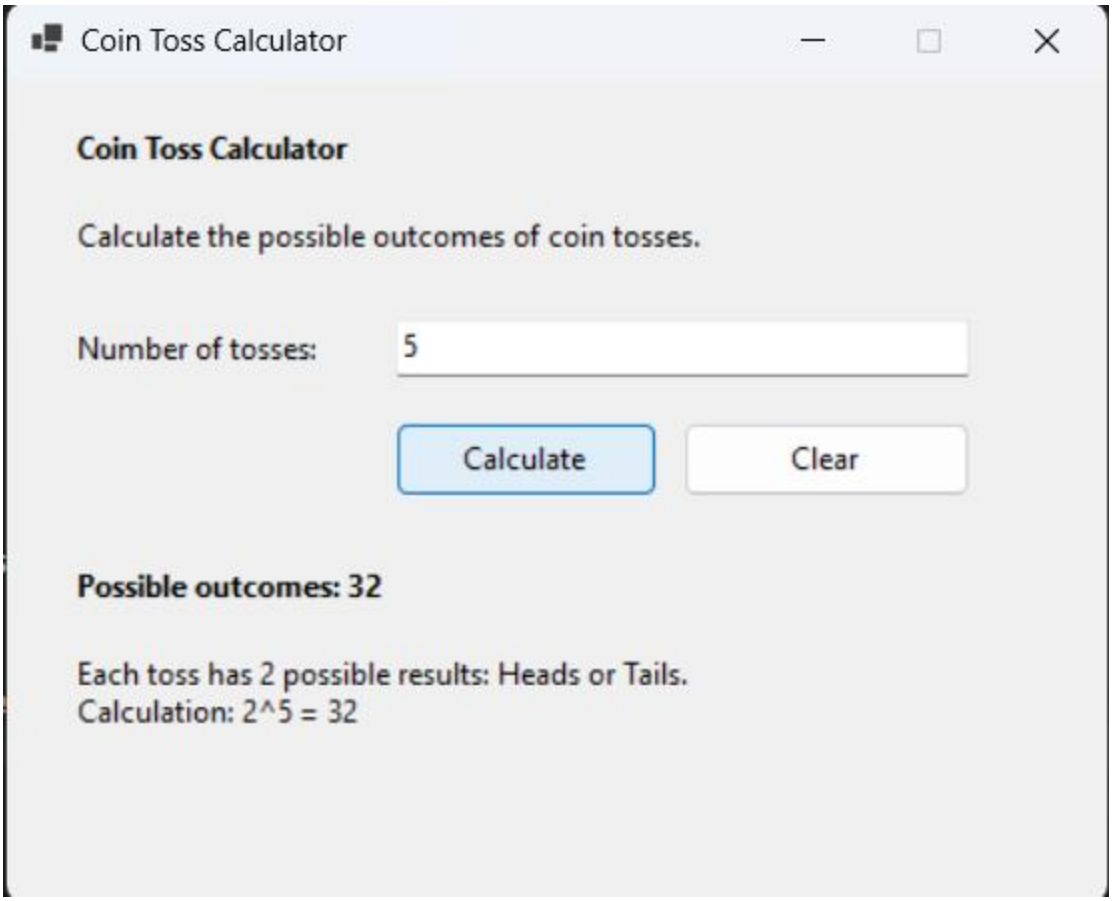


Exponential time

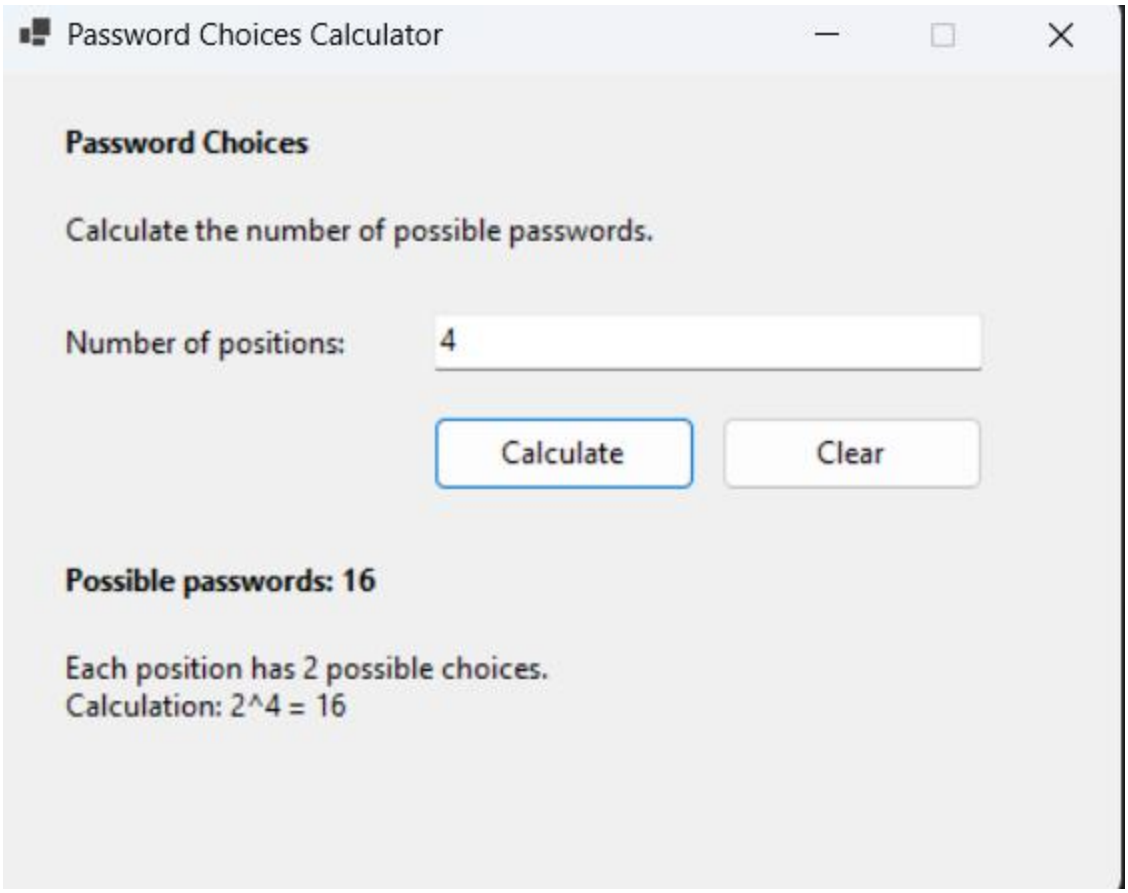
Example1


Light Switches
Light Switches
Click a button to change the switch:
Switch 1: OFF
Switch 2: OFF
Switch 3: OFF
Calculate
Clear
Number of combinations:
8

Example2


Coin Toss Calculator
Coin Toss Calculator
Calculate the possible outcomes of coin tosses.
Number of tosses:
5
Calculate
Clear
Possible outcomes: 32
Each toss has 2 possible results: Heads or Tails.
Calculation: 2^5 = 32

Example3


Password Choices Calculator
Password Choices
Calculate the number of possible passwords.
Number of positions:
4
Calculate
Clear
Possible passwords: 16
Each position has 2 possible choices.
Calculation: 2^4 = 16